

Matrices: Formula Sheet & Reference

Class 12 Mathematics – Chapter 3 Complete Reference & Formulas

1 Definition, Order & Special Types of Matrices

- **Matrix Array Notation:**

$$A = [a_{ij}]_{m \times n} \quad (1 \leq i \leq m, 1 \leq j \leq n)$$

Has m rows, n columns, and mn total elements.

- **1. Column Matrix:** Order $m \times 1$ ($n = 1$).
- **2. Row Matrix:** Order $1 \times n$ ($m = 1$).
- **3. Square Matrix:** $m = n$ (Order $n \times n$).
- **4. Diagonal Matrix:** Square matrix with $a_{ij} = 0$ for all $i \neq j$.
- **5. Scalar Matrix:** Diagonal matrix with equal diagonal entries:

$$a_{ij} = \begin{cases} k, & i = j \\ 0, & i \neq j \end{cases}$$

- **6. Identity Matrix (I_n):** Scalar matrix with $k = 1$:

$$a_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

- **7. Zero Matrix (O):** $a_{ij} = 0$ for all i, j .
- **Equality of Matrices ($A = B$):** Same order ($m \times n$) and $a_{ij} = b_{ij}$ for all i, j .

2 Matrix Operations & Algebraic Properties

- **1. Matrix Addition ($A + B$):** Defined if A and B have identical order $m \times n$:

$$C = A + B \implies c_{ij} = a_{ij} + b_{ij}$$

Properties:

- Commutative: $A + B = B + A$
- Associative: $(A + B) + C = A + (B + C)$
- Additive Identity: $A + O = O + A = A$
- Additive Inverse: $A + (-A) = (-A) + A = O$

- **2. Scalar Multiplication (kA):**

$$kA = [k \cdot a_{ij}]_{m \times n}, \quad -A = (-1)A, \quad A - B = A + (-B)$$

Properties: $k(A + B) = kA + kB$ and $(k + l)A = kA + lA$.

- **3. Matrix Multiplication ($AB = C$):** Defined if columns of A ($m \times n$) equal rows of B ($n \times p$); product C is of order $m \times p$:

$$c_{ik} = \sum_{j=1}^n a_{ij}b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{in}b_{nk}$$

Key Properties:

- Non-Commutative in General: $AB \neq BA$.
- Associative: $(AB)C = A(BC)$.
- Distributive: $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
- Multiplicative Identity: $IA = AI = A$ (for square matrix A).
- Zero Product Rule: $AB = O$ does **not** require $A = O$ or $B = O$.

3 Transpose, Symmetric & Skew-Symmetric Matrices

- **Transpose of a Matrix (A' or A^T):** If $A = [a_{ij}]_{m \times n}$, then $A' = [a_{ji}]_{n \times m}$.
- **Properties of Transpose:**

$$\begin{aligned}(A')' &= A \\ (kA)' &= kA' \\ (A + B)' &= A' + B' \\ (AB)' &= B'A' \quad (\text{Reversal Law})\end{aligned}$$

- **Symmetric Matrix:** A square matrix A is symmetric if:

$$A' = A \iff a_{ji} = a_{ij}, \quad \forall i, j$$

- **Skew-Symmetric Matrix:** A square matrix A is skew-symmetric if:

$$A' = -A \iff a_{ji} = -a_{ij}, \quad \forall i, j$$

Diagonal Element Condition: For $i = j$, $a_{ii} = -a_{ii} \implies 2a_{ii} = 0 \implies a_{ii} = 0$ (All diagonal elements of a skew-symmetric matrix are zero).

- **Theorems on Symmetric and Skew-Symmetric Matrices:**

- **Theorem 1:** For any square matrix A with real entries, $(A + A')$ is symmetric and $(A - A')$ is skew-symmetric.
- **Theorem 2:** Any square matrix A can be uniquely expressed as the sum of a symmetric and a skew-symmetric matrix:

$$A = \underbrace{\frac{1}{2}(A + A')}_{\text{Symmetric Part } P} + \underbrace{\frac{1}{2}(A - A')}_{\text{Skew-Symmetric Part } Q}$$

4 Invertible Matrices & Inversion Properties

- **Definition of Invertible Matrix:** If A is a square matrix of order m , and there exists a square matrix B of order m such that:

$$AB = BA = I$$

Then B is the inverse of A ($B = A^{-1}$), and A is the inverse of B ($A = B^{-1}$).

- **Theorem 3 (Uniqueness of Inverse):** The inverse of a square matrix, if it exists, is unique.
- **Theorem 4 (Reversal Law for Matrix Inversion):** If A and B are invertible matrices of the same order, then:

$$(AB)^{-1} = B^{-1}A^{-1}$$